Angular Momentum Representation For Free particle

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Abstract

This work presents a complete formulation of the angular-momentum representation for a non-relativistic free particle. The central objective is the diagonalization of the Hamiltonian $\hat{H} = \frac{\hat{p}^2}{2m}$ within the basis defined by the Complete Set of Commuting Observables (CSCO): $\{\hat{H}, \hat{L}^2, \hat{L}_z\}$. We explicitly construct the algebraic definition of the angular momentum operator $\hat{L} = \hat{r} \times \hat{p}$, verify its commutation relations with the Hamiltonian, and derive the spherical harmonics $Y_{lm}(\theta, \phi)$ as the coordinate representations of the angular eigenbasis. The study culminates in the derivation of the general solution to the Time-Dependent Schrödinger Equation (TDSE) as a superposition of spherical waves, connecting the spectral evolution to the coordinate representation via the Spherical Bessel transform. Finally, we extend the discussion to the relativistic domain, establishing the link between the Poincaré group, the Dirac equation, and the geometric origin of spin and antiparticles through Clifford algebras.

Keywords: Angular Momentum, Free Particle, Spherical Harmonics, Clifford Algebra, Spin.

1 Problem Setting

We seek a complete formulation of the angular-momentum representation for a non-relativistic free particle. The central objective is the diagonalization of the Hamiltonian

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} \tag{1}$$

within the basis defined by the Complete Set of Commuting Observables (CSCO): $\{\hat{H}, \hat{L}^2, \hat{L}_z\}$. This corresponds to solving the time-dependent Schrödinger equation (TDSE) by exploiting the isotropy of free space.

The formulation is motivated by three theoretical necessities:

1. **Symmetry Realization:** To represent the rotation group SO(3) via its generators acting on the Hilbert space of the free particle.
2. **Conservation Laws:** To explicitly link the invariance of $\hat{H}$ under spatial rotations to the conservation of angular momentum.
3. **Separation of Variables:** To derive the natural basis for spherical geometries, generalizing to central potential problems.

1.1 Physical Scenario

Consider a spinless particle of mass m in Euclidean space $\mathbb{R}^3$. The absence of external potentials implies the Hamiltonian is purely kinetic. The isotropy of the system ensures that the generator of rotations, the angular momentum vector operator $\hat{\mathbf{L}}$, commutes with the Hamiltonian:

$$[\hat{H}, \hat{\mathbf{L}}] = 0. \tag{2}$$

Consequently, stationary states may be labeled simultaneously by energy (E_k), total angular momentum (ℓ), and its projection (m), i.e., $\hat{H}$, $\hat{L}^2$ and $\hat{L}_z$ admit simultaneous eigenstates.

1.2 Construction Objectives

The project proceeds through the following formal steps:

- **Algebraic Definition:** Derivation of the operator $\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}}$ utilizing fundamental canonical commutation relations (CCR).
- **Lie Algebra $\mathfrak{so}(3)$:** Verification of the closure relation $[\hat{L}_i, \hat{L}_j] = i\hbar\,\epsilon_{ijk}\hat{L}_k$, establishing the connection to the structure constants of the rotation group.
- **Spectral Analysis:** Determination of the spectrum for $\hat{L}^2$ and $\hat{L}_z$, yielding the spherical harmonics $Y_{\ell m}(\theta, \phi)$ as coordinate representations of the angular eigenbasis.
- **Basis Transformation:** Construction of the unitary transformation connecting the linear momentum basis $|\mathbf{k}\rangle$ to the spherical basis $|k, \ell, m\rangle$.

1.3 Mathematical Prerequisites

The derivation assumes fluency in:

- **Abstract Formalism:** Hilbert space $\mathcal{H}$, Dirac notation, and spectral theory of Hermitian operators.
- **Symplectic Structure:** Canonical commutators $[\hat{x}_i, \hat{p}_j] = i\hbar\,\delta_{ij}$.
- **Integral Transforms:** Fourier analysis relating position ($\mathbf{r}$) and momentum ($\mathbf{p}$) representations.

- **Special Functions:** Properties of Spherical Harmonics and Spherical Bessel functions.

1.4 Formal Outcome

The construction culminates in the general solution to the TDSE, expressed as a superposition of spherical waves:

$$|\psi(t)\rangle = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \int_0^\infty \mathrm{d}k\, k^2\, \psi_{k\ell m}(0)\, e^{-i\frac{\hbar k^2}{2m}t}\, |k, \ell, m\rangle \,. \tag{3}$$

This expansion connects the abstract eigenkets to the coordinate representation via the spherical Bessel transform.

2 Starting Point: Fundamental Formalism

We formally define the Hilbert space $\mathcal{H} = L^2(\mathbb{R}^3)$ and the dynamical structure of the free particle. The following definitions establish the algebraic and spectral foundations required for the angular-momentum decomposition.

2.1 Dynamics and Unitary Evolution

The dynamics are governed by the self-adjoint Hamiltonian operator $\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m}$. The state vector $|\psi(t)\rangle$ evolves deterministically from an initial state $|\psi_0\rangle$ via the strongly continuous unitary group $\hat{U}(t)$:

$$|\psi(t)\rangle = \hat{U}(t)\, |\psi_0\rangle = \exp\left(-\frac{i}{\hbar}\hat{H}t\right) |\psi_0\rangle \,. \tag{4}$$

This abstract evolution implies the Schrödinger equation $i\hbar\partial_t\, |\psi\rangle = \hat{H}\, |\psi\rangle$ via Stone's Theorem.

2.2 Continuous Representations

We utilize two continuous bases, position $\{|\mathbf{r}\rangle\}$ and linear momentum $\{|\mathbf{p}\rangle\}$, normalized to the Dirac distribution.

2.2.1 Basis Definitions and Orthogonality

The basis vectors satisfy the eigenvalue equations and completeness relations (Resolution of Identity $\hat{\mathbb{I}}$):

$$\hat{\mathbf{r}}\, |\mathbf{r}\rangle = \mathbf{r}\, |\mathbf{r}\rangle \,, \qquad \int_{\mathbb{R}^3} \mathrm{d}^3 r\ |\mathbf{r}\rangle \langle\mathbf{r}| = \hat{\mathbb{I}}, \tag{5}$$

$$\hat{\mathbf{p}}\, |\mathbf{p}\rangle = \mathbf{p}\, |\mathbf{p}\rangle \,, \qquad \int_{\mathbb{R}^3} \mathrm{d}^3 p\ |\mathbf{p}\rangle \langle\mathbf{p}| = \hat{\mathbb{I}}. \tag{6}$$

The inner product defines the transformation kernel (Fourier kernel) between the canonically conjugate variables:

$$\langle \mathbf{r} \rangle \, \mathbf{p} = \frac{1}{(2\pi\hbar)^{3/2}} \exp\left(\frac{i}{\hbar}\mathbf{p}\cdot\mathbf{r}\right). \tag{7}$$

2.2.2 Wavefunction Representations

The projections of the state vector yield the coordinate and momentum space wavefunctions:

- **Coordinate Space:** $\psi(\mathbf{r}, t) \equiv \langle \mathbf{r} \rangle \, \psi(t)$. The momentum operator acts as the generator of spatial translations:

$$\langle \mathbf{r} | \, \hat{\mathbf{p}} \, | \psi \rangle = -i\hbar\nabla\psi(\mathbf{r}). \tag{8}$$

- **Momentum Space:** $\tilde{\psi}(\mathbf{p}, t) \equiv \langle \mathbf{p} \rangle \, \psi(t)$. Since $\hat{H}$ is diagonal in this basis, the time evolution is strictly a phase modulation:

$$\tilde{\psi}(\mathbf{p}, t) = \exp\left(-\frac{i}{\hbar}\frac{p^2}{2m}t\right)\tilde{\psi}(\mathbf{p}, 0). \tag{9}$$

2.3 Spectral Decomposition and Density of States

For the free particle, the energy E is degenerate. The transition from momentum integration to energy integration requires the spectral density function. Using the dispersion relation $E_p = p^2/2m$:

$$\int \mathrm{d}^3 p = \int_0^\infty p^2 \mathrm{d}p \int_{S^2} \mathrm{d}\Omega. \tag{10}$$

Changing variables from p to E introduces the density of states factor via the identity:

$$\mathrm{d}p = \frac{\mathrm{d}p}{\mathrm{d}E}\mathrm{d}E = \frac{m}{p}\mathrm{d}E = \sqrt{\frac{m}{2E}}\mathrm{d}E. \tag{11}$$

This transformation is critical for isolating the radial (energy) dependence from the angular dependence in the final construction.

2.4 Symmetry Generators

The continuous symmetries of the system are generated by Hermitian operators, implying specific conservation laws via Ehrenfest's Theorem, $\frac{\mathrm{d}}{\mathrm{d}t}\langle \hat{A} \rangle = \frac{1}{i\hbar}\langle [\hat{A}, \hat{H}] \rangle$.

1. **Spatial Translations:** Generated by $\hat{\mathbf{p}}$. Since $[\hat{\mathbf{p}}, \hat{H}] = 0$, linear momentum is conserved.

$$\hat{T}(\mathbf{a}) = \exp\left(-\frac{i}{\hbar}\mathbf{a}\cdot\hat{\mathbf{p}}\right). \tag{12}$$

2. **Galilean Boosts:** Generated by $\hat{\mathbf{r}}$ (up to mass scaling). This shifts the momentum spectrum:
$$\hat{T}_{\mathbf{p}}(\mathbf{q}) = \exp\left(\frac{i}{\hbar}\mathbf{q}\cdot\hat{\mathbf{r}}\right). \tag{13}$$

3 The Angular Momentum Operator

In classical mechanics, the angular momentum of a particle with position vector $\mathbf{r}$ and linear momentum $\mathbf{p}$ is defined as the vector product $\mathbf{L}_{cl} = \mathbf{r}\times\mathbf{p}$. To transition to quantum mechanics, we apply the correspondence principle, promoting the dynamical variables to linear operators acting on the Hilbert space: $\mathbf{r}\to\hat{\mathbf{r}}$ and $\mathbf{p}\to\hat{\mathbf{p}}$.

However, since $\hat{\mathbf{r}}$ and $\hat{\mathbf{p}}$ are non-commuting operators, the definition of the vector product must be handled with algebraic rigor to ensure the resulting operator is Hermitian (measurable) and unambiguous.

3.1 Standard Vector Definition

In 3D Euclidean space, we define the operator $\hat{\mathbf{L}}$ as the vector product of the position and momentum operators:

$$\hat{\mathbf{L}} \equiv \hat{\mathbf{r}}\times\hat{\mathbf{p}} = -i\hbar(\mathbf{r}\times\nabla). \tag{14}$$

Using the Levi-Civita pseudotensor ϵ_{ijk}, the components are defined as:

$$\hat{L}_i = \epsilon_{ijk}\hat{x}_j\hat{p}_k, \tag{15}$$

where Einstein summation is implied. This definition ensures that $\hat{\mathbf{L}}$ acts as a vector operator under spatial rotations.

3.2 Geometric Algebra Perspective (Clifford Algebra)

To provide a coordinate-independent definition valid in any dimension, we employ the Geometric Product within the algebra $\mathcal{C}\ell_{3,0}(\mathbb{R})$. The product of the vector operators $\hat{\mathbf{r}}$ and $\hat{\mathbf{p}}$ decomposes into a scalar (symmetric) and bivector (antisymmetric) part:

$$\hat{\mathbf{r}}\hat{\mathbf{p}} = \hat{\mathbf{r}}\cdot\hat{\mathbf{p}} + \hat{\mathbf{r}}\wedge\hat{\mathbf{p}}. \tag{16}$$

The physically significant quantity is the bivector $\hat{\mathcal{L}}$, representing the oriented plane of rotation:
$$\hat{\mathcal{L}} \equiv \hat{\mathbf{r}}\wedge\hat{\mathbf{p}} = \frac{1}{2}(\hat{\mathbf{r}}\hat{\mathbf{p}} - \hat{\mathbf{p}}\hat{\mathbf{r}}). \tag{17}$$
The standard vector operator $\hat{\mathbf{L}}$ is recovered as the Hodge dual of this bivector. Using the unit pseudoscalar $I = \hat{e}_1\hat{e}_2\hat{e}_3$:

$$\hat{\mathbf{L}} = -iI\hat{\mathcal{L}}. \tag{18}$$

This formulation reveals that spin and orbital angular momentum share the same bivector structure, differing only in the vector space they act upon.

3.3 Hermiticity and Observability

For $\hat{\mathbf{L}}$ to represent a physical observable, it must be Hermitian. We examine the adjoint of the component definition (15):

$$\hat{L}_i^\dagger = (\epsilon_{ijk}\hat{x}_j\hat{p}_k)^\dagger. \tag{19}$$

Using the property $(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger\hat{A}^\dagger$ and the fact that $\hat{\mathbf{x}}$ and $\hat{\mathbf{p}}$ are Hermitian ($\hat{x}_j^\dagger = \hat{x}_j$, $\hat{p}_k^\dagger = \hat{p}_k$) and ϵ_{ijk} is real:

$$\hat{L}_i^\dagger = \epsilon_{ijk}\hat{p}_k^\dagger\hat{x}_j^\dagger = \epsilon_{ijk}\hat{p}_k\hat{x}_j. \tag{20}$$

The non-commutativity of $\hat{\mathbf{x}}$ and $\hat{\mathbf{p}}$ is resolved by the canonical commutation relation $[\hat{x}_j, \hat{p}_k] = i\hbar\delta_{jk}$:

$$[\hat{x}_j, \hat{p}_k] = i\hbar\delta_{jk} \implies \hat{p}_k\hat{x}_j = \hat{x}_j\hat{p}_k - i\hbar\delta_{jk}. \tag{21}$$

Substituting this back into the expression for $\hat{L}_i^\dagger$:

$$\begin{aligned}\hat{L}_i^\dagger &= \epsilon_{ijk}(\hat{x}_j\hat{p}_k - i\hbar\delta_{jk})\\ &= \epsilon_{ijk}\hat{x}_j\hat{p}_k - i\hbar\epsilon_{ijk}\delta_{jk}.\end{aligned} \tag{22}$$

The second term contains the contraction of the antisymmetric tensor ϵ_{ijk} with the symmetric Kronecker delta δ_{jk}. Since ϵ_{ijk} vanishes if any two indices are equal (which δ_{jk} forces), we have:

$$\epsilon_{ijk}\delta_{jk} = \epsilon_{ijj} = 0. \tag{23}$$

Therefore:

$$\hat{L}_i^\dagger = \epsilon_{ijk}\hat{x}_j\hat{p}_k = \hat{L}_i. \tag{24}$$

The operator is inherently Hermitian, and no symmetrization (e.g., $\frac{1}{2}(\mathbf{r}\times\mathbf{p} - \mathbf{p}\times\mathbf{r})$) is required because the cross product couples only commuting orthogonal components (e.g., x and p_y).

3.4 Explicit Component Form

Expanding the tensor notation, we obtain the explicit forms used in standard calculations:

$$\hat{L}_x = \hat{y}\hat{p}_z - \hat{z}\hat{p}_y, \tag{25}$$

$$\hat{L}_y = \hat{z}\hat{p}_x - \hat{x}\hat{p}_z, \tag{26}$$

$$\hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x. \tag{27}$$

These expressions confirm that the quantum angular momentum operator preserves the functional form of its classical counterpart, provided the order of operators is maintained as $\mathbf{r}\times\mathbf{p}$.

3.5 The Lie Algebra $\mathfrak{so}(3)$

The fundamental commutation relations for the components of $\hat{\mathbf{L}}$ define the structure of the rotation group. Computing $[\hat{L}_x, \hat{L}_y]$:

$$\begin{aligned}
[\hat{L}_x, \hat{L}_y] &= [yp_z - zp_y, zp_x - xp_z] \\
&= [yp_z, zp_x] + [-zp_y, -xp_z] \quad \text{(other terms commute)} \\
&= yp_x[p_z, z] + xp_y[z, p_z] \\
&= yp_x(-i\hbar) + xp_y(i\hbar) \\
&= i\hbar(xp_y - yp_x) = i\hbar\hat{L}_z.
\end{aligned} \tag{28}$$

Generalizing this result yields the Lie algebra of $\mathfrak{so}(3)$:

$$[\hat{L}_i, \hat{L}_j] = i\hbar\epsilon_{ijk}\hat{L}_k. \tag{29}$$

This closure relation confirms that $\hat{\mathbf{L}}$ generates the group of spatial rotations SO(3).

4 Coordinate Representation of Angular Momentum

We establish the representation of the abstract operator $\hat{\mathbf{L}}$ acting on the Hilbert space of square-integrable functions $L^2(\mathbb{R}^3)$.

4.1 Cartesian Representation

In the position basis $\{|\mathbf{r}\rangle\}$, the momentum operator acts as the gradient generator $\hat{\mathbf{p}} \to -i\hbar\nabla$. Consequently, the angular momentum operator $\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}}$ becomes a differential operator:

$$\hat{\mathbf{L}} = -i\hbar(\mathbf{r} \times \nabla). \tag{30}$$

Expanding the cross product yields the explicit differential forms for the generators of rotations about the Cartesian axes:

$$\begin{aligned}
\hat{L}_x &= -i\hbar\left(y\partial_z - z\partial_y\right), \\
\hat{L}_y &= -i\hbar\left(z\partial_x - x\partial_z\right), \\
\hat{L}_z &= -i\hbar\left(x\partial_y - y\partial_x\right).
\end{aligned} \tag{31}$$

While valid, these coupled expressions obscure the separation of radial and angular dynamics, necessitating a transition to spherical coordinates.

4.2 Spherical Coordinate Transformation

We employ the spherical coordinates (r, θ, ϕ) where $x = r\sin\theta\cos\phi$, $y = r\sin\theta\sin\phi$, and $z = r\cos\theta$.

4.2.1 The Generator of Azimuthal Rotations ($\hat{L}_z$)

The operator $\hat{L}_z$ generates rotations about the z-axis (azimuthal symmetry). Using the chain rule for the partial derivative ∂_ϕ:

$$\frac{\partial}{\partial\phi} = \frac{\partial x}{\partial\phi}\partial_x + \frac{\partial y}{\partial\phi}\partial_y + \frac{\partial z}{\partial\phi}\partial_z = -y\partial_x + x\partial_y. \tag{32}$$

Comparing this to the Cartesian form of $\hat{L}_z$, we identify the identity:

$$\hat{L}_z = -i\hbar\frac{\partial}{\partial\phi}. \tag{33}$$

This simple form confirms that $\hat{L}_z$ is the canonical conjugate momentum to the azimuthal angle ϕ.

To derive the representation of the total angular momentum squared, we utilize the geometric decomposition of the gradient operator. In spherical coordinates, ∇ separates into radial and tangential components:

$$\nabla = \hat{\mathbf{e}}_r\frac{\partial}{\partial r} + \frac{1}{r}\nabla_\Omega, \tag{34}$$

where ∇_Ω denotes the angular gradient on the unit sphere S^2. Substituting this into the definition $\hat{\mathbf{L}} = -i\hbar(\mathbf{r}\times\nabla)$ and noting that $\mathbf{r} = r\hat{\mathbf{e}}_r$:

$$\hat{\mathbf{L}} = -i\hbar\, r\hat{\mathbf{e}}_r \times \left(\hat{\mathbf{e}}_r\partial_r + \frac{1}{r}\nabla_\Omega\right). \tag{35}$$

Since $\hat{\mathbf{e}}_r \times \hat{\mathbf{e}}_r = 0$, the radial derivative vanishes, proving that angular momentum is strictly a transverse operator (independent of radial motion):

$$\hat{\mathbf{L}} = -i\hbar(\hat{\mathbf{e}}_r \times \nabla_\Omega). \tag{36}$$

4.2.2 The Casimir Invariant $\hat{L}^2$

To derive the squared total angular momentum $\hat{L}^2 = \hat{\mathbf{L}}\cdot\hat{\mathbf{L}}$, we avoid the tedious squaring of Cartesian components and instead use the vector properties of the Gradient in spherical coordinates.

The gradient operator ∇ is expressed in the spherical orthonormal basis $(\hat{\mathbf{e}}_r, \hat{\mathbf{e}}_\theta, \hat{\mathbf{e}}_\phi)$ as:

$$\nabla = \hat{\mathbf{e}}_r\frac{\partial}{\partial r} + \hat{\mathbf{e}}_\theta\frac{1}{r}\frac{\partial}{\partial\theta} + \hat{\mathbf{e}}_\phi\frac{1}{r\sin\theta}\frac{\partial}{\partial\phi}. \tag{37}$$

Using $\mathbf{r} = r\hat{\mathbf{e}}_r$ and calculating the cross product $\hat{\mathbf{L}} = -i\hbar(\mathbf{r}\times\nabla)$:

$$\hat{\mathbf{L}} = -i\hbar\left[r\hat{\mathbf{e}}_r \times \left(\hat{\mathbf{e}}_r\frac{\partial}{\partial r} + \hat{\mathbf{e}}_\theta\frac{1}{r}\frac{\partial}{\partial\theta} + \hat{\mathbf{e}}_\phi\frac{1}{r\sin\theta}\frac{\partial}{\partial\phi}\right)\right]. \tag{38}$$

Using the orthogonality relations $\hat{\mathbf{e}}_r \times \hat{\mathbf{e}}_r = 0$, $\hat{\mathbf{e}}_r \times \hat{\mathbf{e}}_\theta = \hat{\mathbf{e}}_\phi$, and $\hat{\mathbf{e}}_r \times \hat{\mathbf{e}}_\phi = -\hat{\mathbf{e}}_\theta$:

$$\begin{aligned}
\hat{\mathbf{L}} &= -i\hbar \left[\left(r\frac{1}{r} \right) (\hat{\mathbf{e}}_r \times \hat{\mathbf{e}}_\theta) \frac{\partial}{\partial \theta} + \left(r \frac{1}{r \sin\theta} \right) (\hat{\mathbf{e}}_r \times \hat{\mathbf{e}}_\phi) \frac{\partial}{\partial \phi} \right] \\
&= -i\hbar \left(\hat{\mathbf{e}}_\phi \frac{\partial}{\partial \theta} - \hat{\mathbf{e}}_\theta \frac{1}{\sin\theta} \frac{\partial}{\partial \phi} \right).
\end{aligned} \tag{39}$$

We apply the vector operator $\hat{\mathbf{L}}$ from Eq. (39) to itself. Extreme care must be taken because the unit vectors $\hat{\mathbf{e}}_\theta$ and $\hat{\mathbf{e}}_\phi$ depend on the coordinates, so derivatives acting on them are non-zero.

$$\hat{L}^2 = (-i\hbar)^2 \left(\hat{\mathbf{e}}_\phi \frac{\partial}{\partial \theta} - \frac{\hat{\mathbf{e}}_\theta}{\sin\theta} \frac{\partial}{\partial \phi} \right) \cdot \left(\hat{\mathbf{e}}_\phi \frac{\partial}{\partial \theta} - \frac{\hat{\mathbf{e}}_\theta}{\sin\theta} \frac{\partial}{\partial \phi} \right). \tag{40}$$

Expanding the dot product:

$$\frac{\hat{L}^2}{-\hbar^2} = \underbrace{\hat{\mathbf{e}}_\phi \cdot \frac{\partial}{\partial \theta} \left(\hat{\mathbf{e}}_\phi \frac{\partial}{\partial \theta} \right)}_{\text{Term 1}} - \underbrace{\hat{\mathbf{e}}_\phi \cdot \frac{\partial}{\partial \theta} \left(\frac{\hat{\mathbf{e}}_\theta}{\sin\theta} \frac{\partial}{\partial \phi} \right)}_{\text{Term 2}} - \underbrace{\frac{\hat{\mathbf{e}}_\theta}{\sin\theta} \cdot \frac{\partial}{\partial \phi} \left(\hat{\mathbf{e}}_\phi \frac{\partial}{\partial \theta} \right)}_{\text{Term 3}} + \underbrace{\frac{\hat{\mathbf{e}}_\theta}{\sin\theta} \cdot \frac{\partial}{\partial \phi} \left(\frac{\hat{\mathbf{e}}_\theta}{\sin\theta} \frac{\partial}{\partial \phi} \right)}_{\text{Term 4}}. \tag{41}$$

Required Derivatives of Unit Vectors:

- $\partial_\theta \hat{\mathbf{e}}_\phi = 0$.
- $\partial_\theta \hat{\mathbf{e}}_\theta = -\hat{\mathbf{e}}_r$ (Orthogonal to $\hat{\mathbf{e}}_\phi$, so dot product vanishes).
- $\partial_\phi \hat{\mathbf{e}}_\phi = -\sin\theta \hat{\mathbf{e}}_r - \cos\theta \hat{\mathbf{e}}_\theta$.
- $\partial_\phi \hat{\mathbf{e}}_\theta = \cos\theta \hat{\mathbf{e}}_\phi$.

Evaluating Terms:
Term 1: Since $\partial_\theta \hat{\mathbf{e}}_\phi = 0$:

$$\hat{\mathbf{e}}_\phi \cdot \left(\hat{\mathbf{e}}_\phi \frac{\partial^2}{\partial \theta^2} \right) = \frac{\partial^2}{\partial \theta^2}.$$

Term 2: Contains $\hat{\mathbf{e}}_\phi \cdot \partial_\theta \hat{\mathbf{e}}_\theta$. Since $\partial_\theta \hat{\mathbf{e}}_\theta = -\hat{\mathbf{e}}_r$, the dot product with $\hat{\mathbf{e}}_\phi$ is 0. Cross terms vanish here.
Term 3:

$$\frac{\hat{\mathbf{e}}_\theta}{\sin\theta} \cdot \left((\partial_\phi \hat{\mathbf{e}}_\phi) \frac{\partial}{\partial \theta} + \hat{\mathbf{e}}_\phi \frac{\partial^2}{\partial \phi \partial \theta} \right).$$

Using $\partial_\phi \hat{\mathbf{e}}_\phi = -\sin\theta \hat{\mathbf{e}}_r - \cos\theta \hat{\mathbf{e}}_\theta$:

$$\frac{\hat{\mathbf{e}}_\theta}{\sin\theta} \cdot \left(-\cos\theta \hat{\mathbf{e}}_\theta \frac{\partial}{\partial \theta} \right) = -\frac{\cos\theta}{\sin\theta} \frac{\partial}{\partial \theta} = -\cot\theta \frac{\partial}{\partial \theta}.$$

Term 4:

$$\frac{\hat{\mathbf{e}}_\theta}{\sin\theta} \cdot \frac{\partial}{\partial \phi} \left(\frac{\hat{\mathbf{e}}_\theta}{\sin\theta} \frac{\partial}{\partial \phi} \right).$$

The derivative acts on $\hat{\mathbf{e}}_\theta$ and the function. However, $\partial_\phi \hat{\mathbf{e}}_\theta = \cos\theta \hat{\mathbf{e}}_\phi$, which is orthogonal to the pre-factor $\hat{\mathbf{e}}_\theta$. Thus, only the derivative on the scalar function survives:

$$\frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}.$$

Summing the non-vanishing contributions:

$$\frac{\hat{L}^2}{-\hbar^2} = \frac{\partial^2}{\partial\theta^2} + \cot\theta\frac{\partial}{\partial\theta} + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}. \tag{42}$$

Recognizing that $\frac{\partial^2}{\partial\theta^2} + \cot\theta\frac{\partial}{\partial\theta} = \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right)$, we arrive at the final expression:

$$\hat{L}^2 = -\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right]. \tag{43}$$

5 Rotational Symmetry and Conservation Laws

We establish the connection between the algebraic generators derived in the previous section and the geometric symmetries of the physical system. This quantizes the classical Noether's theorem: the invariance of the Hamiltonian under spatial rotations implies the conservation of angular momentum.

5.1 The Rotation Operator

5.1.1 Exponential Map

Rotations in the Hilbert space are implemented by the unitary representation of the group SO(3). For a rotation characterized by the vector $\boldsymbol{\theta} = \theta\mathbf{n}$ (rotation by angle θ about axis $\mathbf{n}$), the unitary operator is generated by $\hat{\mathbf{L}}$:

$$\hat{R}(\boldsymbol{\theta}) = \exp\left(-\frac{i}{\hbar}\boldsymbol{\theta}\cdot\hat{\mathbf{L}}\right). \tag{44}$$

Since $\hat{\mathbf{L}}$ is Hermitian, $\hat{R}$ is Unitary ($\hat{R}^\dagger\hat{R} = \mathbb{I}$), preserving the normalization and probability currents of the state.

5.1.2 Transformation of Vector Operators

A vector operator $\hat{\mathbf{V}}$ (such as position $\hat{\mathbf{r}}$ or momentum $\hat{\mathbf{p}}$) is defined by its transformation properties under the adjoint action of the rotation group. Using the Baker-Campbell-Hausdorff lemma, the components transform as:

$$\hat{R}^\dagger(\boldsymbol{\theta})\hat{V}_i\hat{R}(\boldsymbol{\theta}) = \sum_j R_{ij}(\boldsymbol{\theta})\hat{V}_j, \tag{45}$$

where R_{ij} are the components of the classical 3×3 rotation matrix. Specifically for position eigenkets, this implies:

$$\hat{R}(\boldsymbol{\theta}) \left| \mathbf{r} \right\rangle = \left| R\mathbf{r} \right\rangle . \tag{46}$$

5.1.3 Transformation of Scalar Fields

The action of the rotation operator on a scalar wavefunction $\psi(\mathbf{r})$ is derived from the requirement that the physical state remains invariant under a change of frame.

$$\psi'(\mathbf{r}) = \left\langle \mathbf{r} \right\rangle \psi' = \left\langle \mathbf{r} \right| \hat{R}(\boldsymbol{\theta}) \left| \psi \right\rangle . \tag{47}$$

Inserting the identity $\int d^3r' \left| \mathbf{r}' \right\rangle \left\langle \mathbf{r}' \right| = \mathbb{I}$ and using $\left\langle \mathbf{r} \right\rangle \mathbf{r}' = \delta(\mathbf{r} - \mathbf{r}')$:

$$\psi'(\mathbf{r}) = \psi(R^{-1}\mathbf{r}). \tag{48}$$

For an infinitesimal rotation $d\boldsymbol{\theta}$, the inverse rotation is $\mathbf{r} - d\boldsymbol{\theta} \times \mathbf{r}$. Expanding to first order:

$$\begin{aligned}
\psi(\mathbf{r} - d\boldsymbol{\theta} \times \mathbf{r}) &\approx \psi(\mathbf{r}) - (d\boldsymbol{\theta} \times \mathbf{r}) \cdot \nabla\psi(\mathbf{r}) \\
&= \psi(\mathbf{r}) - \frac{i}{\hbar} d\boldsymbol{\theta} \cdot (\mathbf{r} \times -i\hbar\nabla)\psi(\mathbf{r}) \\
&= \left(\mathbb{I} - \frac{i}{\hbar} d\boldsymbol{\theta} \cdot \hat{\mathbf{L}} \right) \psi(\mathbf{r}).
\end{aligned} \tag{49}$$

This recovers the generator definition $\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}}$ consistently from the field transformation perspective.

5.2 Conservation Laws

5.2.1 Scalar Operators

An operator $\hat{S}$ is termed a *scalar operator* if it is invariant under rotations, which implies it commutes with the generators:

$$[\hat{L}_i, \hat{S}] = 0. \tag{50}$$

The dot product of any two vector operators $\hat{\mathbf{A}} \cdot \hat{\mathbf{B}}$ forms a scalar operator. We verify this generally using the tensor commutator $[\hat{L}_i, \hat{A}_j] = i\hbar\epsilon_{ijk}\hat{A}_k$:

$$\begin{aligned}
[\hat{L}_i, \hat{\mathbf{A}} \cdot \hat{\mathbf{B}}] &= [\hat{L}_i, \hat{A}_j \hat{B}_j] \\
&= [\hat{L}_i, \hat{A}_j]\hat{B}_j + \hat{A}_j[\hat{L}_i, \hat{B}_j] \\
&= i\hbar\epsilon_{ijk}\hat{A}_k\hat{B}_j + i\hbar\epsilon_{ijk}\hat{A}_j\hat{B}_k.
\end{aligned} \tag{51}$$

Summing over j, k, the term $\hat{A}_k\hat{B}_j$ is symmetric (assuming $[\hat{A}, \hat{B}] = 0$ or similar vector types) while ϵ_{ijk} is antisymmetric. Thus, the sum vanishes.

5.2.2 Invariance of the Hamiltonian

The free particle Hamiltonian $\hat{H} = \frac{\hat{\mathbf{p}}\cdot\hat{\mathbf{p}}}{2m}$ is a specific instance of a scalar operator (contraction of vector $\hat{\mathbf{p}}$ with itself). Therefore:

$$[\hat{L}_i, \hat{H}] = \frac{1}{2m}[\hat{L}_i, \hat{\mathbf{p}}^2] = 0. \tag{52}$$

5.2.3 Time Independence

Applying the Heisenberg equation of motion (or Ehrenfest's theorem):

$$\frac{d}{dt}\hat{\mathbf{L}}_H(t) = \frac{1}{i\hbar}[\hat{\mathbf{L}}, \hat{H}] + \frac{\partial \hat{\mathbf{L}}}{\partial t} = 0. \tag{53}$$

This confirms that for an isotropic free particle, angular momentum is a strict constant of motion, and the states may be simultaneously labeled by energy and angular momentum quantum numbers.

5.2.4 Classical Correspondence and Isomorphism

The conservation law established above is the quantum manifestation of the classical rotational invariance. Classically, the time evolution of angular momentum is governed by the Poisson bracket with the Hamiltonian:

$$\frac{d\mathbf{L}_{cl}}{dt} = \{\mathbf{L}_{cl}, H_{cl}\}_{PB}. \tag{54}$$

For a free particle, the absence of external torque implies $\{\mathbf{L}_{cl}, H_{cl}\}_{PB} = 0$. The quantum result is structurally identical, following the Dirac quantization correspondence:

$$\lim_{\hbar \to 0} \frac{1}{i\hbar}[\hat{\mathbf{L}}, \hat{H}] \longrightarrow \{\mathbf{L}_{cl}, H_{cl}\}_{PB}. \tag{55}$$

This confirms that the role of $\hat{\mathbf{L}}$ is twofold and consistent across regimes: it is simultaneously the physical observable corresponding to classical angular momentum and the infinitesimal generator of the rotation group SO(3) imposed by the isotropy of the vacuum.

6 The Lie Algebra of Rotations

The quantization of angular momentum is structurally defined by the non-commutative geometry of the rotation group SO(3). We derive the Lie algebra $\mathfrak{so}(3)$ directly from the canonical commutation relations (CCR) of position and momentum.

6.1 Fundamental Commutation Relations

We compute the commutator $[\hat{L}_x, \hat{L}_y]$ using the operator definition $\hat{L}_i = \epsilon_{ijk}\hat{x}_j\hat{p}_k$ and the Heisenberg algebra $[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}$.

6.1.1 Algebraic Derivation

Expanding the components $\hat{L}_x = \hat{y}\hat{p}_z - \hat{z}\hat{p}_y$ and $\hat{L}_y = \hat{z}\hat{p}_x - \hat{x}\hat{p}_z$:

$$[\hat{L}_x, \hat{L}_y] = [\hat{y}\hat{p}_z - \hat{z}\hat{p}_y, \hat{z}\hat{p}_x - \hat{x}\hat{p}_z]. \tag{56}$$

Using the linearity of the commutator and the product rule $[AB, C] = A[B, C]+[A, C]B$, we identify the non-vanishing terms. The only non-commuting pairs involve coordinates and momenta of the same index (specifically z and p_z here).

$$\begin{aligned}
[\hat{L}_x, \hat{L}_y] &= [\hat{y}\hat{p}_z, \hat{z}\hat{p}_x] + [-\hat{z}\hat{p}_y, -\hat{x}\hat{p}_z] \quad \text{(Cross terms vanish)} \\
&= \hat{y}[\hat{p}_z, \hat{z}]\hat{p}_x + \hat{x}\hat{p}_y[\hat{z}, \hat{p}_z] \\
&= \hat{y}(-i\hbar)\hat{p}_x + \hat{x}\hat{p}_y(i\hbar) \\
&= i\hbar(\hat{x}\hat{p}_y - \hat{y}\hat{p}_x) = i\hbar\hat{L}_z.
\end{aligned} \tag{57}$$

Cyclic permutation of indices ($x \to y \to z$) yields the complete Lie algebra of the group:

$$[\hat{L}_i, \hat{L}_j] = i\hbar\epsilon_{ijk}\hat{L}_k. \tag{58}$$

6.1.2 Geometric Interpretation

The non-commutativity $[\hat{L}_x, \hat{L}_y] \neq 0$ reflects the non-Abelian nature of 3D rotations: a rotation about the x-axis followed by the y-axis is distinct from the reverse order. In the language of Geometric Algebra, this arises because the generators are bivectors (oriented planes); the plane xy (generated by L_z) is not orthogonal to the interaction of planes yz and zx.

6.2 Commutation of the Casimir Invariant $\hat{L}^2$

The total angular momentum operator is defined as the contraction of the vector operator with itself:

$$\hat{L}^2 \equiv \hat{\mathbf{L}} \cdot \hat{\mathbf{L}} = \sum_i \hat{L}_i^2. \tag{59}$$

This operator is a scalar (rank-0 tensor) and represents the Casimir invariant of the $\mathfrak{so}(3)$ algebra.

6.2.1 Commutation with Generators

To verify that $\hat{L}^2$ commutes with the generators, we compute $[\hat{L}^2, \hat{L}_k]$. Using the identity $[\hat{A}^2, \hat{B}] = \hat{A}[\hat{A}, \hat{B}] + [\hat{A}, \hat{B}]\hat{A}$:

$$\begin{aligned}
[\hat{L}^2, \hat{L}_z] &= [\hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2, \hat{L}_z] \\
&= [\hat{L}_x^2, \hat{L}_z] + [\hat{L}_y^2, \hat{L}_z] \quad (\text{since } [\hat{L}_z, \hat{L}_z] = 0) \\
&= \hat{L}_x[\hat{L}_x, \hat{L}_z] + [\hat{L}_x, \hat{L}_z]\hat{L}_x + \hat{L}_y[\hat{L}_y, \hat{L}_z] + [\hat{L}_y, \hat{L}_z]\hat{L}_y.
\end{aligned} \tag{60}$$

Substituting the structure constants from Eq. (58):

- $[\hat{L}_x, \hat{L}_z] = -i\hbar\hat{L}_y$
- $[\hat{L}_y, \hat{L}_z] = +i\hbar\hat{L}_x$

$$
\begin{aligned}
[\hat{L}^2, \hat{L}_z] &= -i\hbar(\hat{L}_x\hat{L}_y + \hat{L}_y\hat{L}_x) + i\hbar(\hat{L}_y\hat{L}_x + \hat{L}_x\hat{L}_y) \\
&= 0. \qquad (61)
\end{aligned}
$$

By symmetry, $[\hat{L}^2, \hat{\mathbf{L}}] = 0$.

6.2.2 Physical Significance: Compatible Observables

The commutation $[\hat{L}^2, \hat{L}_z] = 0$ implies that the magnitude of rotation (related to the radius of the sphere) and the plane of rotation (orientation) are compatible physical properties.

- Scalar Nature: $\hat{L}^2$ is a scalar operator; it measures the "amount" of rotation invariant under coordinate changes.
- Vector Nature: $\hat{L}_z$ is a component of a vector (or bivector) and depends on the choice of axis.

Because they commute, they share a common basis of eigenstates $|\ell, m\rangle$, allowing for the simultaneous quantization of total energy, total angular momentum, and azimuthal orientation.

7 Eigenvalue Problem of $\hat{L}^2$ and $\hat{L}_z$

The commutation $[\hat{L}^2, \hat{L}_z] = 0$ implies the existence of a common eigenbasis for the Hilbert space of functions on the unit sphere $L^2(S^2)$. We seek the simultaneous eigenfunctions $Y(\theta, \phi)$ satisfying:

$$\hat{L}_z Y(\theta, \phi) = \lambda Y(\theta, \phi), \qquad (62)$$

$$\hat{L}^2 Y(\theta, \phi) = \mu Y(\theta, \phi). \qquad (63)$$

7.1 Spectrum of $\hat{L}_z$: Azimuthal Quantization

The operator $\hat{L}_z = -i\hbar\partial_\phi$ depends solely on the azimuthal angle. Separation of variables $Y(\theta, \phi) = f(\theta)\Phi(\phi)$ yields the differential equation:

$$\frac{d\Phi}{d\phi} = \frac{i\lambda}{\hbar}\Phi(\phi). \qquad (64)$$

The general solution is the complex exponential $\Phi(\phi) = Ae^{i(\lambda/\hbar)\phi}$.

7.1.1 Topological Constraint

The domain of ϕ is the circle S^1. For the wavefunction to be single-valued (a necessary condition for the probability density to be physically well-defined), it must satisfy periodic boundary conditions:

$$\Phi(\phi + 2\pi) = \Phi(\phi) \implies e^{i(\lambda/\hbar)2\pi} = 1. \tag{65}$$

This condition restricts the spectrum of $\hat{L}_z$ to discrete values:

$$\lambda_m = m\hbar, \quad m \in \mathbb{Z}. \tag{66}$$

Thus, the magnetic quantum number m arises from the rotational symmetry of the system and the topology of the configuration space.

7.2 Spectrum of $\hat{L}^2$: Polar Quantization

Using the eigenvalue λ_m and the spherical representation of $\hat{L}^2$, the second eigenvalue equation becomes:

$$-\hbar^2 \left[\frac{1}{\sin\theta} \partial_\theta (\sin\theta \partial_\theta) - \frac{m^2}{\sin^2\theta} \right] f(\theta) = \mu f(\theta). \tag{67}$$

Letting $\mu = \hbar^2 \nu$, this simplifies to the generalized Legendre equation.

7.2.1 Transformation to the Legendre Equation

Introducing the variable $x = \cos\theta$ (mapping $\theta \in [0, \pi] \to x \in [-1, 1]$), the equation transforms to:

$$\frac{d}{dx} \left[(1 - x^2) \frac{df}{dx} \right] + \left(\nu - \frac{m^2}{1 - x^2} \right) f(x) = 0. \tag{68}$$

7.2.2 Regularity Condition

The differential equation (68) has regular singular points at the poles $x = \pm 1$. A Frobenius series analysis reveals that the general solution diverges logarithmically at these boundaries unless the series terminates. The condition for termination (and thus square-integrability) is that the eigenvalue parameter ν must take the form:

$$\nu = \ell(\ell + 1), \quad \ell \in \mathbb{Z}_{\geq 0}. \tag{69}$$

This quantizes the magnitude of angular momentum:

$$\mu_\ell = \hbar^2 \ell(\ell + 1). \tag{70}$$

7.3 The Spherical Harmonics $Y_{\ell m}$

The non-singular solutions to the angular equation are the Associated Legendre Polynomials $P_\ell^m(x)$.

$$Y_{\ell m}(\theta, \phi) = (-1)^m \sqrt{\frac{2\ell+1}{4\pi}\frac{(\ell-m)!}{(\ell+m)!}} P_\ell^m(\cos\theta) e^{im\phi}. \tag{71}$$

The phase factor $(-1)^m$ is the Condon-Shortley phase convention.

7.3.1 Constraints on Quantum Numbers

The structure of the solution space imposes the constraint $|m| \le \ell$. This can be understood via the Rodrigues formula for the polynomials:

$$P_\ell^m(x) \propto (1-x^2)^{m/2} \frac{d^{\ell+m}}{dx^{\ell+m}} (x^2-1)^\ell. \tag{72}$$

Since $(x^2-1)^\ell$ is a polynomial of degree 2ℓ, derivatives of order higher than 2ℓ vanish. However, a more subtle constraint arises from the definition of the angular momentum ladder operators $\hat{L}_\pm$. If $|m| > \ell$, the state would have a norm less than zero, which is impossible in a Hilbert space. Thus, for a fixed ℓ:

$$m \in \{-\ell, -\ell+1, \ldots, \ell-1, \ell\}. \tag{73}$$

7.4 Summary of the Angular Basis

We have constructed a complete orthonormal basis $|\ell, m\rangle$ for the angular variables. The defining actions of the operators are:

$$\hat{L}^2 |\ell, m\rangle = \hbar^2 \ell(\ell+1) |\ell, m\rangle, \tag{74}$$

$$\hat{L}_z |\ell, m\rangle = \hbar m |\ell, m\rangle. \tag{75}$$

These states provide the necessary separation of variables to solve the radial part of the Schrödinger equation for the free particle.

8 Dynamics and the Radial Wave Equation

We formally solve the Time-Dependent Schrödinger Equation (TDSE) by projecting the dynamics onto the eigenbasis of the Complete Set of Commuting Observables (CSCO) $\{\hat{H}, \hat{L}^2, \hat{L}_z\}$. This procedure separates the trivial temporal evolution from the geometric spatial structure.

8.1 Spectral Evolution

The state vector $|\psi(t)\rangle$ is expanded in the continuous basis of simultaneous eigenstates $|k, \ell, m\rangle$, normalized to the radial measure $k^2\mathrm{d}k$:

$$|\psi(t)\rangle = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \int_0^{\infty} \mathrm{d}k\, k^2\, c_{k\ell m}(t)\ |k, \ell, m\rangle\,. \tag{76}$$

Substituting this expansion into the TDSE $i\hbar\partial_t |\psi\rangle = \hat{H} |\psi\rangle$ and exploiting the orthogonality of the basis, the partial differential equation decouples into an infinite set of independent ordinary differential equations for the coefficients $c_{k\ell m}(t)$:

$$i\hbar\dot{c}_{k\ell m}(t) = E_k c_{k\ell m}(t), \quad \text{where } E_k = \frac{\hbar^2 k^2}{2m}. \tag{77}$$

The solution is exact and demonstrates that the angular quantum numbers are constants of motion:

$$c_{k\ell m}(t) = c_{k\ell m}(0) \exp\left(-\frac{i}{\hbar}\frac{\hbar^2 k^2}{2m} t\right). \tag{78}$$

8.2 The Coordinate Representation (Radial Equation)

To determine the spatial probability distribution, we must find the coordinate representation of the basis kets, $\langle \mathbf{r}|k, \ell, m\rangle$. We exploit the separation of variables in spherical coordinates:

$$\langle \mathbf{r}|k, \ell, m\rangle = R_{k\ell}(r) Y_{\ell m}(\theta, \phi). \tag{79}$$

Here, $Y_{\ell m}$ are the Spherical Harmonics derived previously. The radial function $R_{k\ell}(r)$ is determined by the energy eigenvalue equation in position space:

$$\langle \mathbf{r}|\hat{H}|k, \ell, m\rangle = E_k \langle \mathbf{r}|k, \ell, m\rangle\,. \tag{80}$$

Using the Laplacian decomposition $\nabla^2 = \partial_r^2 + \frac{2}{r}\partial_r - \frac{\hat{L}^2}{\hbar^2 r^2}$, the equation becomes:

$$-\frac{\hbar^2}{2m}\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial R_{k\ell}}{\partial r}\right) - \frac{\ell(\ell+1)}{r^2} R_{k\ell}(r)\right] = \frac{\hbar^2 k^2}{2m} R_{k\ell}(r). \tag{81}$$

8.2.1 The Spherical Bessel Equation

Rearranging terms and introducing the dimensionless variable $\rho = kr$, we obtain the Spherical Bessel differential equation:

$$\frac{d^2 R}{d\rho^2} + \frac{2}{\rho}\frac{dR}{d\rho} + \left[1 - \frac{\ell(\ell+1)}{\rho^2}\right] R(\rho) = 0. \tag{82}$$

The term $\frac{\ell(\ell+1)}{\rho^2}$ represents the centrifugal potential barrier arising from angular momentum conservation, which prevents high-ℓ particles from penetrating near the origin.

8.2.2 Free Spherical Waves

The physically admissible solutions (regular at the origin $r = 0$) are the Spherical Bessel Functions of the first kind, denoted $j_\ell(kr)$.

$$R_{k\ell}(r) = \sqrt{\frac{2}{\pi}} j_\ell(kr). \tag{83}$$

(The normalization factor $\sqrt{2/\pi}$ is chosen to satisfy the delta-function normalization $\int r^2 dr R_{k\ell} R_{k'\ell} = \delta(k - k')/k^2$).

8.3 The Full Wavefunction Solution

Combining the spectral evolution with the coordinate basis, the general solution for the free particle in the angular momentum representation is:

$$\psi(\mathbf{r}, t) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \int_0^\infty \mathrm{d}k\, k^2 \underbrace{c_{k\ell m}(0) e^{-iE_k t/\hbar}}_{\text{Dynamics}} \underbrace{\sqrt{\frac{2}{\pi}} j_\ell(kr) Y_{\ell m}(\theta, \phi)}_{\text{Spatial Mode}}. \tag{84}$$

This expansion represents the free particle as a superposition of spherical standing waves, providing the natural basis for scattering problems where the asymptotic behavior is defined by incoming and outgoing spherical states.

9 Time Evolution and General Solution

The diagonalization of the Hamiltonian within the CSCO basis $\{|k, \ell, m\rangle\}$ decouples the global evolution of the field $\psi(\mathbf{r}, t)$ into an infinite set of independent dynamical modes. We construct the general solution to the initial value problem by applying the unitary time-evolution operator to the spectral expansion.

9.1 Unitary Evolution of Spectral Modes

The equation of motion derived implies that each spectral amplitude $\psi_{k\ell m}(t)$ undergoes simple harmonic oscillation in the complex plane. The formal solution is obtained via the action of the propagator $\hat{U}(t) = \exp(-i\hat{H}t/\hbar)$:

$$\psi_{k\ell m}(t) = \langle k, \ell, m | \hat{U}(t) | \psi(0) \rangle = e^{-iE_k t/\hbar} \psi_{k\ell m}(0). \tag{85}$$

Explicitly, with the free particle dispersion relation $E_k = \frac{\hbar^2 k^2}{2m}$:

$$\psi_{k\ell m}(t) = \psi_{k\ell m}(0) \exp\left(-i \frac{\hbar k^2}{2m} t \right). \tag{86}$$

This result highlights the stationarity of the angular momentum eigenstates; the probability density $|\psi_{k\ell m}(t)|^2$ is time-invariant, ensuring that information encoded in the angular channels is conserved.

9.2 Synthesis of the Dynamical State

The full dynamical state $|\psi(t)\rangle$ is reconstructed by synthesizing the evolved modes. Substituting the spectral solution (86) into the basis expansion yields the general solution in abstract notation:

$$|\psi(t)\rangle = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \int_0^{\infty} \mathrm{d}k\, k^2\, \psi_{k\ell m}(0)\, e^{-i\frac{\hbar k^2}{2m}t}\; |k, \ell, m\rangle\,. \tag{87}$$

Projecting this state onto the coordinate basis $\langle \mathbf{r}|$ and utilizing the free spherical wave functions $\langle \mathbf{r}\rangle\, k, \ell, m = \sqrt{\frac{2}{\pi}} j_\ell(kr) Y_{\ell m}(\Omega)$, we obtain the explicit position-space wavefunction:

$$\psi(\mathbf{r}, t) = \sqrt{\frac{2}{\pi}} \sum_{\ell,m} Y_{\ell m}(\Omega) \int_0^{\infty} \mathrm{d}k\, k^2\, \psi_{k\ell m}(0)\, j_\ell(kr)\, e^{-i\frac{\hbar k^2}{2m}t}. \tag{88}$$

This integral transform constitutes the complete solution to the free-particle Schrödinger equation in spherical coordinates.

9.3 Physical Interpretation

The structure of Eq. (88) elucidates the physical mechanism of free evolution:

- **Angular Stability:** The quantum numbers ℓ and m appear strictly as summation indices for the stationary harmonics $Y_{\ell m}$. The absence of time dependence in the angular sector confirms that the probability distribution over the orbital angular momentum channels is invariant, a direct consequence of the isotropy of free space ($[\hat{H}, \hat{\mathbf{L}}] = 0$).
- **Radial Dispersion:** Time dependence is confined to the phase modulation of the radial modes. Since the dispersion relation is non-linear ($E_k \propto k^2$), the group velocity $v_g = \partial_k \omega = \hbar k/m$ varies with wavenumber. This gradient in propagation speed across different k-components drives the characteristic wave packet spreading (dispersion) observed in the position representation.

Ultimately, the angular momentum representation resolves the complex diffusive dynamics of the free particle into a static rotation of independent spectral components in the complex plane.

10 Integral Transforms and Representations

The abstract solution $|\psi(t)\rangle$ acquires physical meaning only when projected onto specific basis manifolds. We construct the explicit integral transforms that map the angular momentum spectral amplitudes $\psi_{k\ell m}$ to the observable representations of position, momentum, and energy.

10.1 Position Space: The Spherical Bessel Transform

The transition to coordinate space is defined by the transformation kernel $\langle \mathbf{r}|k,\ell,m\rangle$. This kernel arises from the partial-wave expansion of the plane wave, identifying the free spherical wave as the radial eigenfunction.

10.1.1 The Transformation Kernel

We adopt the normalization convention consistent with the quantum Fourier transform:

$$\langle \mathbf{r}|k,\ell,m\rangle = i^\ell \sqrt{\frac{2}{\pi}} j_\ell(kr) Y_{\ell m}(\Omega_r). \tag{89}$$

The phase factor i^ℓ is essential for time-reversal consistency.

10.1.2 Wavefunction Synthesis

Projecting the spectral expansion Eq. (87) onto the position bra $\langle \mathbf{r}|$ yields the spatial field:

$$\psi(\mathbf{r},t) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} i^\ell Y_{\ell m}(\Omega_r) \int_0^\infty \mathrm{d}k\, k^2 \underbrace{\sqrt{\frac{2}{\pi}} j_\ell(kr)}_{\text{Radial Basis}} \psi_{k\ell m}(t). \tag{90}$$

This expression identifies the radial wavefunction $R_\ell(r,t)$ as the Spherical Bessel Transform (or Hankel transform of order $\ell + 1/2$) of the spectral amplitude $\psi_{k\ell m}$. This is the spherical analog to the standard Fourier transform, mapping the radial momentum domain k to the radial position domain r.

10.2 Momentum Space: Harmonic Analysis on the Sphere

In the linear momentum representation, the physics is simplified by the fact that $\hat{\mathbf{p}}$ commutes with the free Hamiltonian. The angular momentum states define a specific geometric partitioning of momentum space.

10.2.1 Diagonal Representation

The basis states $|k,\ell,m\rangle$ are eigenstates of the momentum magnitude operator $\hat{p} = \sqrt{\hat{\mathbf{p}}^2}$. Consequently, the projection kernel is sharply localized (distributional):

$$\langle \mathbf{p}|k,\ell,m\rangle = \frac{\delta(p - \hbar k)}{p^2} Y_{\ell m}(\Omega_p). \tag{91}$$

Here, Ω_p denotes the angles of the momentum vector.

10.2.2 Momentum Distribution

Substituting this kernel into the general solution collapses the radial integral immediately:

$$\begin{aligned}\psi(\mathbf{p},t) &= \int_0^\infty \mathrm{d}k\, k^2\, \psi_{k\ell m}(t) \left[\frac{\delta(p-\hbar k)}{p^2} Y_{\ell m}(\Omega_p)\right] \\ &= \frac{1}{\hbar^3}\psi_{p/\hbar,\ell,m}(t) Y_{\ell m}(\Omega_p). \end{aligned} \tag{92}$$

This result confirms that the spectral coefficients $\psi_{k\ell m}$ are physically equivalent to the momentum-space wavefunction projected onto spherical harmonics. The time evolution is strictly a phase rotation, as momentum states are stationary solutions of the free particle.

10.3 Energy Representation and Density of States

The energy representation is crucial for scattering theory (S-matrix formalism). Since the spectrum of the free particle is continuous and degenerate, the mapping involves the Density of States (DOS).

10.3.1 Jacobian Transformation

Using the dispersion relation $E = \frac{\hbar^2 k^2}{2m}$, we relate the differential measures:

$$\mathrm{d}k = \frac{\mathrm{d}k}{\mathrm{d}E}\mathrm{d}E = \frac{1}{\hbar}\sqrt{\frac{m}{2E}}\mathrm{d}E. \tag{93}$$

The density of states factor is $\rho(E) \propto \sqrt{E}$.

10.3.2 Spectral Evolution

The wavefunction in the energy representation $\psi(E,\ell,m)$ is related to the k-space amplitude by preserving the probability norm $\int |\psi|^2 \mathrm{d}k = \int |\phi|^2 \mathrm{d}E$:

$$\psi(E,\ell,m;t) = \left(\frac{mk(E)}{\hbar^2}\right)^{1/2} \psi_{k(E),\ell,m}(0) e^{-iEt/\hbar}. \tag{94}$$

This representation trivializes the dynamics to a linear phase evolution, explicitly demonstrating that the angular momentum modes are the stationary states of the system.

11 Synthesis of the Non-Relativistic Formalism

The construction of the angular momentum representation for the free particle serves as a complete laboratory for the axioms of non-relativistic quantum mechanics. We conclude by abstracting the mathematical structure into a set of formal postulates and defining the geometric decomposition of the state space.

11.1 Fundamental Postulates

Based on the spectral analysis of the operators $\hat{\mathbf{r}}$, $\hat{\mathbf{p}}$, $\hat{\mathbf{L}}$, and $\hat{H}$, we encapsulate the theory in four structural definitions:

1. **The State Space:** The physical state of the system is represented by a ray in a complex, separable Hilbert space $\mathcal{H} \cong L^2(\mathbb{R}^3)$.
2. **Observables:** Measurable physical quantities correspond to self-adjoint (Hermitian) operators acting on $\mathcal{H}$. The possible outcomes of a measurement are restricted to the spectrum $\sigma(\hat{A})$ of the operator.
3. **Born Rule (Spectral Projection):** For a system in state $|\psi\rangle$, the probability of measuring an eigenvalue $\lambda \in \sigma(\hat{A})$ is given by the expectation value of the projection operator $\hat{P}_\lambda$:

$$P(\lambda) = \langle\psi|\hat{P}_\lambda\psi\rangle = |\langle\lambda|\psi\rangle|^2. \tag{95}$$

4. **Unitary Evolution:** The time evolution of a closed system is deterministic and generated by the Hamiltonian $\hat{H}$, forming a strongly continuous unitary group $\hat{U}(t) = e^{-i\hat{H}t/\hbar}$.

11.2 Hilbert Space Decomposition and the CSCO

The rotational symmetry of the free Hamiltonian implies a specific geometric structure for the Hilbert space. We utilize the Complete Set of Commuting Observables (CSCO) $\mathcal{S} = \{\hat{H}, \hat{L}^2, \hat{L}_z\}$ to partition the space.

11.2.1 Tensor Product Structure

The Hilbert space factorizes into a radial component and an angular component:

$$\mathcal{H} \cong \mathcal{H}_{\text{radial}} \otimes \mathcal{H}_{\text{angular}} = L^2(\mathbb{R}^+, r^2 dr) \otimes L^2(S^2, d\Omega). \tag{96}$$

11.2.2 Invariant Subspaces

Under the action of the rotation group SO(3), the angular space decomposes into a direct sum of orthogonal invariant subspaces $\mathcal{E}^{(\ell)}$, each of dimension $2\ell + 1$. The total Hilbert space is the direct sum:

$$\mathcal{H} = \bigoplus_{\ell=0}^{\infty} \mathcal{H}^{(\ell)}. \tag{97}$$

Each subspace $\mathcal{H}^{(\ell)}$ contains states with definite total angular momentum ℓ, which remain invariant in magnitude under rotations, mixing only among the m-components.

11.2.3 The Symmetry-Adapted Eigenbasis

The basis vectors spanning these subspaces are the simultaneous eigenvectors of the CSCO, denoted $|E, \ell, m\rangle$ (or $|k, \ell, m\rangle$):

$$\hat{H} |k, \ell, m\rangle = \frac{\hbar^2 k^2}{2m} |k, \ell, m\rangle \,, \tag{98}$$

$$\hat{L}^2 |k, \ell, m\rangle = \hbar^2 \ell(\ell + 1) |k, \ell, m\rangle \,, \tag{99}$$

$$\hat{L}_z |k, \ell, m\rangle = \hbar m |k, \ell, m\rangle \,. \tag{100}$$

This basis diagonalizes the dynamics, reducing the 3D Schrödinger PDE to a 1D radial ODE for each ℓ-channel. The Completeness Relation (Resolution of Identity) for this decomposition is:

$$\hat{\mathbb{I}} = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \int_0^{\infty} \mathrm{d}k \; |k, \ell, m\rangle \langle k, \ell, m| \,. \tag{101}$$

This summation explicitly reconstructs the full space from the irreducible representations of the symmetry group.

11.3 Symmetries and Conservation Laws (Quantum Noether's Theorem)

The derivation of the angular momentum eigenvalues exhibits the profound link between continuous symmetries and conservation laws. In the quantum regime, Noether's theorem is realized through the algebraic properties of the unitary generators of symmetry groups.

11.3.1 Invariance and Generators

Let $\mathcal{G}$ be a continuous group of transformations acting on the Hilbert space, parameterized by a real variable α and generated by the self-adjoint operator $\hat{G}$:

$$\hat{U}(\alpha) = \exp\left(-\frac{i}{\hbar}\alpha\hat{G}\right). \tag{102}$$

The system is invariant under this symmetry if the Hamiltonian commutes with the unitary transformation, $[\hat{H}, \hat{U}(\alpha)] = 0$, which for infinitesimal transformations implies:

$$[\hat{H}, \hat{G}] = 0. \tag{103}$$

By the Heisenberg equation of motion, this commutation ensures that the generator $\hat{G}$ is a constant of motion:

$$\frac{d}{dt}\hat{G}_H(t) = \frac{1}{i\hbar}[\hat{G}_H, \hat{H}] = 0. \tag{104}$$

11.3.2 Symmetries of the Free Particle

For the free particle Hamiltonian $\hat{H} = \hat{\mathbf{p}}^2/2m$, the maximal symmetry group of Euclidean space dictates the fundamental conservation laws:

- **Spatial Homogeneity (Translation Group** T_3**):** Invariant under $\mathbf{r} \to \mathbf{r} + \mathbf{a}$.

$$[\hat{H}, \hat{\mathbf{p}}] = 0 \implies \frac{d\langle \hat{\mathbf{p}} \rangle}{dt} = 0 \quad \text{(Linear Momentum)}. \tag{105}$$

- **Spatial Isotropy (Rotation Group** $\mathrm{SO}(3)$**):** Invariant under $\mathbf{r} \to \mathcal{R}\mathbf{r}$.

$$[\hat{H}, \hat{\mathbf{L}}] = 0 \implies \frac{d\langle \hat{\mathbf{L}} \rangle}{dt} = 0 \quad \text{(Angular Momentum)}. \tag{106}$$

- **Time Homogeneity:** Invariant under $t \to t + \tau$.

$$[\hat{H}, \hat{H}] = 0 \implies \frac{d\langle \hat{H} \rangle}{dt} = 0 \quad \text{(Energy)}. \tag{107}$$

11.3.3 Classical Correspondence: The Poisson Bracket

The structural unity between classical and quantum mechanics is established via the Dirac quantization rule. Classically, the time evolution of an observable A is governed by the Poisson bracket with the Hamiltonian:

$$\frac{dA}{dt} = \{A, H\}_{PB} + \frac{\partial A}{\partial t}. \tag{108}$$

The quantum commutator is the algebraic isomer of the Poisson bracket, scaled by the deformation parameter $\hbar$. In the limit $\hbar \to 0$, we recover the classical law:

$$\lim_{\hbar \to 0} \frac{1}{i\hbar}[\hat{A}, \hat{H}] \longrightarrow \{A, H\}_{PB}. \tag{109}$$

Consequently, the quantum statement $[\hat{L}_i, \hat{H}] = 0$ corresponds directly to the classical statement $\{L_i, H\}_{PB} = 0$. This isomorphism ensures that the conservation of angular momentum is a robust feature of the central force problem, valid across both microscopic and macroscopic regimes.

11.4 Classical-Quantum Isomorphism and Formalism

The structural correspondence between the classical symplectic manifold and the quantum Hilbert space is preserved via the Dirac quantization map. This mapping is not merely a syntactic substitution but represents a functorial transition between two distinct mathematical categories.

11.4.1 The Dynamical Dictionary

The following dictionary establishes the algebraic isomorphism between the commutative algebra of classical functions and the non-commutative algebra of quantum operators:

Table 1 Structural Correspondence: Geometry vs. Algebra

Concept	Classical Mechanics (Geometric)	Quantum Mechanics (Algebraic)
Arena	Symplectic Manifold $\mathcal{M} \cong T^*\mathbb{Q}$	Complex Hilbert Space $\mathcal{H}$
State	Point $\xi = (\mathbf{r}, \mathbf{p}) \in \mathcal{M}$	Ray $\|\psi\rangle \in \mathbb{P}\mathcal{H}$
Observables	Real Functions $C^\infty(\mathcal{M})$	Hermitian Operators $\mathcal{B}_{sa}(\mathcal{H})$
Structure	Commutative, Associative	Non-Commutative, Associative
Lie Bracket	Poisson Bracket $\{A, B\}_{PB}$	Commutator $\frac{1}{i\hbar}[\hat{A}, \hat{B}]$
Generator	Hamiltonian Vector Field X_H	Hamiltonian Operator $\hat{H}$
Dynamics	$\frac{df}{dt} = \{f, H\}_{PB}$	$\frac{d\hat{A}}{dt} = \frac{1}{i\hbar}[\hat{A}, \hat{H}]$

11.4.2 From Differential Geometry to Algebraic Topology

This dictionary highlights a fundamental shift in mathematical formalism:

- **Classical Mechanics as Differential Geometry:** Classical dynamics is the study of Differential Topology and Symplectic Geometry. The state is a point on a smooth differentiable manifold. Time evolution is a continuous flow generated by a vector field tangent to the manifold. The algebra of observables is commutative (order of multiplication does not matter), reflecting the geometric commutativity of coordinate functions.
- **Quantum Mechanics as Algebraic Geometry:** Quantum mechanics abandons the point-set topology of the manifold in favor of Non-Commutative Geometry (Algebraic Topology). The uncertainty principle $[\hat{x}, \hat{p}] \neq 0$ eliminates the concept of a phase space "point" Instead, the theory is defined by the algebraic structure of the operators themselves (C^*-algebra). The geometry is "spectral", encoded in the eigenvalues and eigenstates of the algebra rather than spatial coordinates.

11.5 The Unifying Role of Clifford Algebras

While the formalisms appear distinct, Clifford Algebra (Geometric Algebra) provides the bridge that elucidates the geometric origin of algebraic quantum structures.

1. **The Bivector Nature of Rotation:** In both classical and quantum mechanics, angular momentum is fundamentally a bivector quantity, $\mathbf{L} \sim \mathbf{r} \wedge \mathbf{p}$.
   - In the differential geometry of CM, this bivector generates flows along closed curves on the manifold.

- In the algebraic geometry of QM, this bivector appears in the commutation relations of the generators of the Lie algebra $\mathfrak{so}(3)$.

2. **Spin and Topology:** Differential geometry on SO(3) captures integer angular momentum ($\ell \in \mathbb{Z}$). However, the algebraic structure of Clifford algebras naturally reveals the Spinor representations (half-integer spin) which arise from the simply connected covering group $\mathrm{SU}(2) \cong \mathrm{Spin}(3)$.

   Thus, Clifford algebras demonstrate that the "algebraic" quirks of quantum mechanics (like non-commutativity and spin) are actually deep "geometric" properties of spacetime that classical differential geometry fails to resolve.

12 Relativistic Quantum Mechanics and the Origin of Spin

12.1 The Poincaré Group and Particle Representations

The unification of special relativity with quantum mechanics necessitates that physical states transform consistently under the symmetries of spacetime. The full isometry group of Minkowski spacetime is the Poincaré group, $\mathrm{IO}(1,3) \cong \mathrm{T}(1,3) \rtimes \mathrm{O}(1,3)$, which combines spacetime translations with Lorentz transformations. Within the Wigner-Bargmann framework, elementary particles are defined as irreducible unitary representations of this group.

While the Klein-Gordon equation corresponds to the scalar representation (spin-0), carrying the trivial representation of the Lorentz group, this is insufficient to describe matter particles (fermions). Fermions require a projective (spinor) representation, specifically the $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$ representation, to account for their intrinsic angular momentum. Consequently, a deep analysis of relativistic wave equations is equivalent to studying the fundamental representation theory of spacetime symmetries.

12.2 Derivation of the Dirac Equation

The derivation begins with the relativistic Hamiltonian for a free particle:

$$H = \sqrt{p^2c^2 + m^2c^4}. \tag{110}$$

Direct application of the correspondence principle ($E \to i\hbar\partial_t$, $p \to -i\hbar\nabla$) to this Hamiltonian results in a wave equation containing a square-root operator, which acts as a non-local pseudo-differential operator. To avoid this, one typically squares the energy relation to $E^2 = p^2c^2 + m^2c^4$, yielding the Klein-Gordon equation. However, this second-order equation leads to non-positive definite probability densities and fails to describe spin-1/2 particles.

To resolve these issues, Dirac insisted on an equation that was first-order in the time derivative (like Schrödinger's) to ensure a positive-definite probability density. Relativistic covariance then demands that the equation must also be first-order in spatial derivatives. Dirac proposed a first-order Lorentz-covariant operator $D = \gamma_\mu \partial_\mu$ that "squares" to the d'Alembertian operator of the Klein-Gordon equation ($D^2 =$

$c^{-2}\partial_t^2 - \nabla^2$). This requirement leads to the Dirac equation:

$$i\hbar \left(\gamma_0 \frac{1}{c} \frac{\partial}{\partial t} + \gamma_1 \frac{\partial}{\partial x_1} + \gamma_2 \frac{\partial}{\partial x_2} + \gamma_3 \frac{\partial}{\partial x_3} \right) \psi = mc\psi. \tag{111}$$

12.3 Encoding Spin and Antimatter

A relativistic first-order equation is impossible if the coefficients are scalars. For the operator D to factorize the Laplacian, the coefficients γ_μ must satisfy the defining anticommutation relations of the Clifford algebra $\mathcal{C}l_{1,3}(\mathbb{R})$:

$$\{\gamma_\mu, \gamma_\nu\} = \gamma_\mu\gamma_\nu + \gamma_\nu\gamma_\mu = 2\eta_{\mu\nu}\mathbb{I}. \tag{112}$$

This algebraic structure dictates that the wavefunction ψ cannot be a scalar but must be a spinor residing in a representation space of the algebra. Thus, spin is not an ad hoc addition but an automatic consequence of linearizing the relativistic energy relation; the equation naturally encodes the magnetic moment with gyromagnetic ratio $g = 2$. Furthermore, this structure necessitates the existence of charge-conjugate solutions, predicting the existence of antimatter (the positron).

12.4 Bosons, Fermions, and Statistics

The distinction between bosons and fermions arises from their behavior under rotations. Lorentz transformations are realized as inner automorphisms of the Clifford algebra acting via spin groups, which provide a covering of the $\mathrm{SO}^+(3,1)$ group. While bosonic fields (like those obeying the Klein-Gordon equation) transform under the standard vector or scalar representations ($\mathrm{SO}(3)$), fermionic fields transform under spinor representations ($\mathrm{SU}(2)$), acquiring a phase factor of -1 under a 2π rotation.

This geometric distinction induces the statistical nature of the particles. Understanding the Clifford algebra is essential to grasp the distinction between the Commutative algebras governing Bosonic fields and the Grassmann-graded algebras required for Fermionic matter. These anticommutation relations constitute the finite-dimensional archetype of the Canonical Anticommutation Relations (CAR) in Quantum Field Theory, leading directly to the antisymmetry of multiparticle wavefunctions and the Pauli exclusion principle.

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